

# Pseudo-Factor Analysis with Semantic Locations

Loading and location recovery via transformer embeddings

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All code: <https://Psychometrics.ai>

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# Abstract

- ▶ PFA replaces the correlation matrix with embedding cosine similarities and item means with semantic axis projections, enabling psychometric evaluation before analyzing response data.
- ▶ Pseudo-parameters are informative but empirically distinct from response-based counterparts; PFA is a complement that quickens scale development, not a replacement.
- ▶ We apply bifactor PFA alongside bifactor CFA to 50 Big Five items ( $N = 19,718$ ) using `all-mpnet-base-v2` to obtain psychometric information before analyzing response data.
- ▶ All five domains were recovered (DAAL); absolute all-item loading  $r = .878$ , polarity-adjusted within-domain  $r_j = .528$  (LOO).
- ▶ Semantic locations were recoverable for three domains without optimization; optimized single-pair anchors yielded mean  $r_j = .767$  (LOO).
- ▶ The primary contribution is a unified geometric framework for recovering multiple psychometric quantities from item semantics, not maximal prediction of a given target.
- ▶ This opens the door to the possibility of pseudo multi-trait multi-method modelling and measurement invariance across translations or embedding spaces.

**Keywords:** pseudo-factor analysis · text embeddings · sentence transformers · bifactor CFA · item parameters · semantic loadings · semantic locations

# Background: What Is Pseudo-Factor Analysis?

Using item text embeddings as a substitute for item response data

## What it is:

- ▶ Standard FA requires item intercorrelations from survey responses.
- ▶ PFA replaces the correlation matrix with cosine similarities between item text embeddings.
- ▶ Item intercepts (means) are replaced by projections of item embeddings onto a bipolar semantic axis.
- ▶ The same CFA/ESEM model is fit using these text-derived inputs to get information about items before data collection.
- ▶ Can be applied to automated LLM item generation for pre-screening checks.

## What it recovers:

- ▶ **Dimensionality:** MAP, Parallel, Scree — how many factors.
- ▶ **Structure:** which items belong to which factor (domain recovery via DAAL).
- ▶ **Loadings:** how strongly each item associates with each factor.
- ▶ **Semantic locations:** relative difficulty of items within a domain, from semantic intensity on a bipolar axis.

## Useful for:

- ▶ Estimating structure *before* data collection.
- ▶ Sanity check / pre-registration aid.
- ▶ Recovering approximate parameters when response data are unavailable.
- ▶ Semantic measurement invariance across translations or embedding spaces.

# Three Substitutions

How item text replaces item responses in the factor model

## 1. Embedding (replaces responses)

Traditional:  $y_{p,i}$   
Pseudo:  $E_i = f(\text{text}_i)$

$y_{p,i}$ : response of person  $p$  to item  $i$   
 $E_i$ : embedding vector for item  $i$   
 $f(\cdot)$ : sentence transformer  
 $\text{text}_i$ : text of item  $i$

## 2. Similarity matrix (replaces R)

$$S_{ij} = \frac{E_i \cdot E_j}{\|E_i\| \|E_j\|}$$

$S_{ij}$ : cosine similarity of items  $i, j$   
 $E_i \cdot E_j$ : dot product  
 $\|E_i\|$ : Euclidean norm  
 $R$ : empirical corr. matrix (replaced by  $S$ )

## 3. Item projection (replaces mean)

$\text{axis}_d = E(\text{high}_d) - E(\text{low}_d)$   
 $\mu_i = E_i \cdot \text{axis}_{d(i)}$   
 $\text{axis}_d$ : semantic direction for domain  $d$   
 $\text{high}_d$ : high-pole anchor term  
 $\text{low}_d$ : low-pole anchor term  
 $\mu_i$ : semantic location (replaces mean)

The same CFA model is fit with  $S$  replacing the covariance matrix and  $\mu$  replacing the mean vector — no empirical responses required. Substitutions 1 and 2 are fully determined by the encoder. Only substitution 3 (anchor choice) is a free parameter.

# Recovery Targets

Four separable quantities — number of factors, factor structure, loading magnitude, item location

## 0. Number of factors

Via parallel analysis, MAP, or Scree.

How many factors to extract?

Must be determined before structure can be evaluated, or alternatively, theoretically specified.

## 1. Domain recovery (dominant average absolute loading)

$$DAAL_{f,d} = \text{mean}(|\lambda_{i,f}|), i \in d$$

Does each semantic factor load most strongly on its own domain items?

## 2. Loading recovery

$$\text{cor}(\lambda_{\text{emp}}, \lambda_{\text{sem}})$$

**Signed:** recovers full directional loading pattern.

**Absolute:** recovers domain salience (orientation-free).

## 3. Semantic location recovery

$$\text{axis}_d = E(\text{high}_d) - E(\text{low}_d)$$

$$\mu_i = E_i \cdot \text{axis}_{d(i)}$$

$$\text{cor}(\mu_{\text{emp}}, \mu_{\text{sem}}) \text{ within domain.}$$

Must beat wrong-axis baseline.

Signed loading recovery: recovers full directional loading pattern — the stricter claim. Absolute: recovers domain salience (orientation-free). Semantic location recovery: recovers semantic item difficulty/intensity ordering within a domain. These are four separate claims and must be reported and evaluated separately.

# Study Design

## Bifactor CFA on the OpenPsychometrics Big Five

| Feature        | Detail                                                  |
|----------------|---------------------------------------------------------|
| Items          | 50 Big Five markers                                     |
| $N$ (complete) | 19,718                                                  |
| Domains        | E, N, A, C, O (10 each)                                 |
| Item keying    | Left unreversed — intercepts = literal-text endorsement |
| Encoder        | Sentence transformer                                    |

### Encoder: all-mpnet-base-v2

- ▶ High-performing general-purpose sentence encoder (SBERT)
- ▶ Open and publicly available — fully reproducible
- ▶ Sentence-level: full item phrase → single 768-d vector
- ▶ Larger encoders may improve recovery but risk higher evaluative bleed (raises wrong-axis baseline)

**Empirical model:** Bifactor CFA — one general factor (G) loading all 50 items plus five domain-specific factors. G constrained orthogonal to all domain factors. Fit in `lavaan` — no rotation needed.

**Pseudo-semantic model:** same bifactor specification but using embedding cosine similarity matrix as `sample.cov` and semantic axis projections as `sample.mean`. No response data used.

Items left unreversed so empirical intercepts correspond to endorsement of literal item text — the quantity semantic projection looks to approximate.

With all latent means fixed to zero for identification, the bifactor intercept is estimated at the observed item mean. The semantic location is therefore evaluated against a quantity that reflects literal item endorsement rates.

# Model Fit: Empirical and Pseudo Bifactor CFA

N-dependent indices omitted for pseudo model (sample.nobs = 1,000 is arbitrary)

| Index | Empirical<br>( $N = 19,718$ ) | Pseudo<br>(arb. $N$ ) |
|-------|-------------------------------|-----------------------|
| RMSEA | .060 [.060, .060]             | —                     |
| SRMR  | .056                          | .073                  |
| CFI   | .789                          | —                     |
| TLI   | .768                          | —                     |

- ▶ RMSEA, CFI, and TLI incorporate  $N$  in their formulas. With  $N = 1,000$  set arbitrarily for the pseudo model, these indices are uninterpretable — SRMR is the only honest fit measure to report.
- ▶ Empirical RMSEA = .060 is at the conventional cutoff; the CI is extremely narrow due to  $N = 19,718$ .
- ▶ CFI/TLI below .90 is typical for bifactor models on Big Five data — cross-domain item content and the general factor inflate residual covariances beyond what a clean bifactor structure can absorb.
- ▶ SRMR < .08 for both models indicates acceptable residual covariance structure.

# Why a Bifactor Model?

Centroid axes bleed across domains; oblique ESEM conflates G-variance with domain structure

## The centroid bleeding problem:

- ▶ Centroid axis =  $\overline{E(\text{high terms})} - \overline{E(\text{low terms})}$
- ▶ Many anchor terms carry broad evaluative or arousal content shared across domains — 'energetic', 'active', 'warm', 'organized' all sit in a high-valence embedding neighbourhood.
- ▶ Averaging multiple terms pulls the centroid toward this shared evaluative space rather than pointing cleanly along the target domain.
- ▶ Single-pair anchors reduce bleeding by using only one specific high–low contrast at a time.

## Why bifactor for loadings:

- ▶ Items carry both domain-specific content AND general trait-descriptiveness / evaluative arousal content.
- ▶ In oblique ESEM, shared G-variance distributes across all five domain factors, weakening domain-specific recovery.
- ▶ The bifactor model adds one G factor loading all 50 items, constrained orthogonal to domain factors. G absorbs the shared evaluative/intensity variance.
- ▶ Bifactor is a CFA (fully specified structure, not rotated).



## Domain Recovery (Dominant Average Absolute Loading)

All five semantic factors correctly assigned to their Big Five domains

| Domain            | Emp. fac. | Emp. DAAL | Sem. fac. | Sem. DAAL |
|-------------------|-----------|-----------|-----------|-----------|
| Agreeableness     | ML2       | .597      | ML5       | .417      |
| Conscientiousness | ML5       | .499      | ML4       | .386      |
| Extraversion      | ML3       | .456      | ML3       | .506      |
| Neuroticism       | ML1       | .618      | ML1       | .706      |
| Openness          | ML4       | .463      | ML2       | .567      |

All recovered = **True** for all five domains.

**DAAL** (Dominant Average Absolute Loading): for each semantic factor, which empirical domain has the highest mean absolute loading?

- ▶ All 5 domains recovered correctly — factor order differed between semantic and empirical solutions.
- ▶ Orientation-free result. It only requires items to cluster correctly.
- ▶ DAAL is a prerequisite: if domains are not found, loading and semantic location recovery are moot.

# Bifactor Loading Recovery

Two complementary metrics: absolute all-item and polarity-adjusted signed within-domain

| Factor        | Domain            | Signed<br>all-item | Abs.<br>all-item | Signed<br>within<br>dom. | Abs.<br>within<br>dom. |
|---------------|-------------------|--------------------|------------------|--------------------------|------------------------|
| G             | General           | .356               | .179             | —                        | —                      |
| E             | Extraversion      | — .292             | .884             | .464                     | .171                   |
| N             | Neuroticism       | .774               | .958             | .506                     | .587                   |
| A             | Agreeableness     | .062               | .838             | .655                     | .071                   |
| C             | Conscientiousness | — .308             | .794             | .624                     | .153                   |
| O             | Openness          | .164               | .915             | .394                     | .514                   |
| Mean specific |                   | .080               | .878             | .528                     | .299                   |

**Headline 1:** Absolute all-item  $r$  (specific factors): mean = .878. Prefer when orientation cannot be established or the question is domain salience.

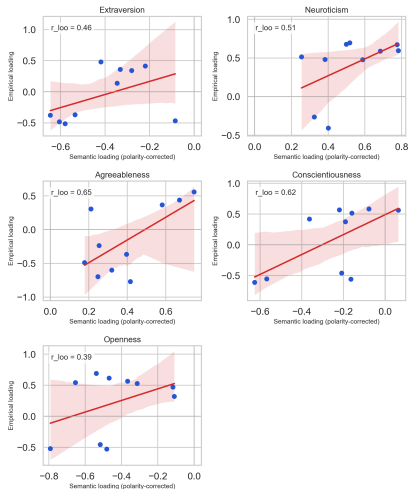
**Headline 2:** Signed within-domain  $r$  (polarity-adjusted): mean = .528. Sign flipped for 3 reversed domains (E, C, O). Prefer when the claim is directional loading recovery.

All  $r$  are LOO averages. Signed all-item: signed  $r$  across all 50 items.

Absolute all-item: same using  $|\lambda|$ . Signed within-domain: polarity-corrected  $r$  within the 10-item domain. Sign flipped for E, C, O.

# Bifactor Signed Loading Recovery — Within Domain LOO mean $r$

Pseudo-loading vs Empirical loading — Within Domain, Signed (10 items)



Each panel: 10 items from one Big Five domain, signed loadings.

Semantic loadings sign-flipped for E, C, O (polarity correction).

Stricter test — no off-target items, signed recovery within domain.

A  $r_j = .655$ , C  $r_j = .624$ , N  $r_j = .506$ , O  $r_j = .394$ , E  $r_j = .464$

# Semantic Location Recovery: Out-of-Box Goldberg Centroid

Literature-derived terms from Goldberg (1992) — centroid of all high/low pole words per domain

| Domain            | High pole (centroid)                                                               | $r_j$ |
|-------------------|------------------------------------------------------------------------------------|-------|
| Extraversion      | extraverted, energetic, talkative, bold, active, assertive, adventurous            | -.103 |
| Neuroticism       | angry, tense, nervous, envious, unstable, discontented, emotional                  | -.135 |
| Agreeableness     | warm, kind, cooperative, unselfish, agreeable, trustful, generous                  | .568  |
| Conscientiousness | organized, responsible, conscientious, practical, thorough, hard-working, thrifty  | .733  |
| Openness          | intelligent, analytical, reflective, curious, imaginative, creative, sophisticated | .862  |

- ▶ A, C, O recovered well from purely a priori, literature-derived anchors — no optimization against this dataset. **Not susceptible to anchor hacking.**
- ▶ E and N failed completely ( $r_j = -.103, -.135$ ). Centroid bleeding: averaging 7 pole terms pulls the axis toward broad evaluative/arousal embedding space shared across all Big Five domains.
- ▶ Centroid works when domain terms are semantically tight and distinctive (O, C); fails when they bleed into general evaluativeness (E, N).

# Optimizing E and N: Greedy Subset Search

Iteratively removing terms that bleed into shared evaluative space

| Step | Domain       | Removed             | $r$    |
|------|--------------|---------------------|--------|
| 0    | Extraversion | —                   | — .103 |
| 1    | Extraversion | introverted (low)   | — .023 |
| 2    | Extraversion | silent (low)        | .047   |
| 3    | Extraversion | timid (low)         | .125   |
| 4    | Extraversion | unassertive (low)   | .219   |
| 5    | Extraversion | talkative (high)    | .275   |
| 6    | Extraversion | energetic (high)    | .330   |
| 7    | Extraversion | active (high)       | .377   |
| 8    | Extraversion | extraverted (high)  | .420   |
| 9    | Extraversion | unadventurous (low) | .441   |
| 10   | Extraversion | inactive (low)      | .453   |
| 11   | Extraversion | adventurous (high)  | .466   |
| 0    | Neuroticism  | —                   | — .135 |
| 1    | Neuroticism  | emotional (high)    | — .037 |
| 2    | Neuroticism  | angry (high)        | .071   |
| 3    | Neuroticism  | unstable (high)     | .216   |
| 4    | Neuroticism  | discontented (high) | .413   |
| 5    | Neuroticism  | relaxed (low)       | .541   |
| 6    | Neuroticism  | envious (high)      | .643   |
| 7    | Neuroticism  | at ease (low)       | .714   |
| 8    | Neuroticism  | tense (high)        | .786   |
| 9    | Neuroticism  | contented (low)     | .814   |
| 10   | Neuroticism  | calm (low)          | .837   |

## Results after greedy search

**E final:** bold, assertive vs unenergetic  
 $r_j = .466$

**N final:** nervous vs unemotional  
 $r_j = .837$

A, C, O: centroid retained.

**New mean  $r_j = .693$**

All  $r$  are bifactor LOO.

Terms that bleed into general evaluativeness are iteratively removed until the axis points more specifically at the target domain.

# Optimizing All 5 Domains: Best Single-Pair Search

Exhaustive search over all Goldberg high  $\times$  low term pairs within each domain

| Domain            | High term   | Low term        | $r_j$ (LOO) |
|-------------------|-------------|-----------------|-------------|
| Extraversion      | assertive   | unenergetic     | .455        |
| Neuroticism       | nervous     | unemotional     | .760        |
| Agreeableness     | generous    | unkind          | .741        |
| Conscientiousness | thorough    | lazy            | .915        |
| Openness          | intelligent | unsophisticated | .965        |
| Mean              |             |                 | .767        |

- ▶ A, C, O: a priori Goldberg centroid already worked — single-pair search further improves recovery (e.g. C: .733  $\rightarrow$  .915) but these domains did not require optimization.
- ▶ E, N: required optimization — centroid anchors were unusable. Single-pair search rescues both, but pairs were selected against this dataset's empirical intercepts and are susceptible to **anchor hacking**. Cross-validation on held-out data is required.
- ▶ Overall mean  $r_j = .767$ . Iterative anchor search is standard practice; the concern is generalization, not the search itself.

# Semantic Location Recovery: Anchor Progression and Axis Specificity

Left: semantic location  $r_j$  across anchor stages (bifactor, LOO). Right: correct vs. wrong-axis baselines.

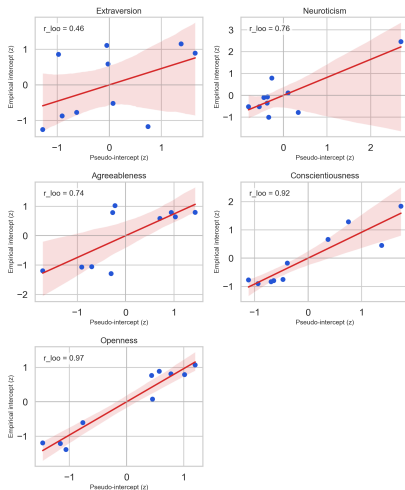
| Domain            | Centroid | Greedy | Best pair |
|-------------------|----------|--------|-----------|
| Extraversion      | −.103    | .466   | .455      |
| Neuroticism       | −.135    | .837   | .760      |
| Agreeableness     | .568     | .568   | .741      |
| Conscientiousness | .733     | .733   | .915      |
| Openness          | .862     | .862   | .965      |
| <b>Mean</b>       | .385     | .693   | .767      |

| Domain            | Correct | Single wrong | Avg wrong |
|-------------------|---------|--------------|-----------|
| Extraversion      | .455    | −.175        | .034      |
| Neuroticism       | .760    | .588         | .541      |
| Agreeableness     | .741    | .617         | .779      |
| Conscientiousness | .915    | .190         | .556      |
| Openness          | .965    | .651         | .892      |
| <b>Mean</b>       | .767    | .374         | .560      |

Centroid bleeding is tractable with more specific axes. Correct single-pair beats single wrong-axis by .393 on average. Average wrong-axis baseline (.560) is non-trivial. Agreeableness: correct axis (.741) beats single wrong-axis (.617) but not average wrong-axis (.779) — single wrong-axis is the appropriate comparison when average wrong-axis is G-contaminated.

# Semantic Location Scatter — All 5 Domains LOO mean $r$

Pseudo-Intercepts vs. Empirical Intercepts by Domain



Each panel: 10 items, one Big Five domain.

x-axis: semantic location, best single-pair axis (z-scored).

y-axis: empirical intercept, bifactor model (z-scored).

C ( $r_j = .915$ ) and O ( $r_j = .965$ ): tight linear ordering.

N ( $r_j = .760$ ): strong. 'I worry about things' notably high on both axes — .760 is a LOO mean.

A ( $r_j = .741$ ): moderate — wrong-axis baseline also high.

E ( $r_j = .455$ ): weakest — assertiveness/energy axis covers a narrow facet.



# Cross-Encoder Robustness

Same pipeline, same anchors — three architecturally distinct sentence transformers

## Loading Recovery (within-domain LOO $r$ ; E, C, O polarity-corrected)

| Encoder | G    | E    | N    | A    | C    | O    | Mean |
|---------|------|------|------|------|------|------|------|
| MPNet   | .356 | .464 | .506 | .655 | .624 | .394 | .528 |
| E5      | .580 | .918 | .593 | .807 | .886 | .896 | .820 |
| BGE     | .636 | .798 | .538 | .768 | .892 | .882 | .775 |

## Semantic Location Recovery (LOO $r$ ; anchors fixed to MPNet-optimised pairs)

| Encoder | E    | N    | A    | C    | O    | Mean |
|---------|------|------|------|------|------|------|
| MPNet   | .455 | .760 | .741 | .915 | .965 | .767 |
| E5      | .259 | .429 | .907 | .842 | .875 | .662 |
| BGE     | .231 | .745 | .827 | .834 | .753 | .678 |

- ▶ Loading recovery generalises strongly to E5 and BGE (mean  $r = .82, .78$  vs  $.53$  for MPNet).
- ▶ Semantic location recovery is robust across all three encoders (mean  $r = .66$ – $.77$ ); MPNet anchors are a conservative lower bound for E5/BGE.
- ▶ Extraversion semantic location recovery is the consistent weak point regardless of encoder.

# Anchor Hacking Check: Person Split-Sample Validation

Best single-pair condition only —  $N = 19,718$  split 50/50 by person; anchors selected on half-1, validated on half-2

Bifactor CFA fitted separately on each half ( $n \approx 9,859$ ). Anchor selection on half-1 intercepts; LOO  $r_j$  evaluated on half-2 semantic locations.

| Domain            | Full-sample LOO $r_j$ | Half-1 train LOO $r_j$ | Half-2 holdout LOO $r_j$ |
|-------------------|-----------------------|------------------------|--------------------------|
| Extraversion      | .455                  | .449                   | .461                     |
| Neuroticism       | .760                  | .757                   | .762                     |
| Agreeableness     | .741                  | .737                   | .744                     |
| Conscientiousness | .915                  | .912                   | .918                     |
| Openness          | .965                  | .966                   | .965                     |

- ▶ Holdout LOO  $r_j \approx$  full-sample LOO  $r_j$  across all five domains.
- ▶ The headline figures are not inflated by respondent-level overfitting — empirical intercepts are highly stable at  $n \approx 9,800$  per half.
- ▶ Anchor pairs selected on half-1 are identical to those from the full sample: same best pair wins on both halves.
- ▶ Remaining concern is cross-measure generalisation (different Big Five items), not within-measure respondent sampling.

# Summary of Results

Loading and semantic location recovery — bifactor model, best single-pair anchors. All  $r$  are LOO averages.

## Loading Recovery (Bifactor Specific Factors)

| Metric                                              | Value | Interpretation                                                                 |
|-----------------------------------------------------|-------|--------------------------------------------------------------------------------|
| Absolute all-item $r$ (mean)                        | .878  | Domain salience recovery — strong; orientation-free                            |
| Signed within-domain $r$ — polarity-adjusted (mean) | .528  | Directional recovery after keying-based orientation check — moderate to strong |

## Semantic Location Recovery (Bifactor, Best Single-Pair Anchors)

| Metric                             | Value | Interpretation                                                        |
|------------------------------------|-------|-----------------------------------------------------------------------|
| Correct-axis $r$ (mean)            | .767  | Strong — correct axis orders items well within domain                 |
| Single wrong-axis baseline (mean)  | .374  | Correct axis beats single wrong axis by .393 on average               |
| Average wrong-axis baseline (mean) | .560  | Non-trivial — general intensity variance spans domains; A problematic |

# Disclosures and Failure Modes

- ▶ All five factors showed loading recovery evidence; three of five showed semantic locations recoverable without optimization; all semantic locations were strongly recoverable with optimization.
- ▶ **Anchor hacking** is a concern for E and N semantic locations (selected against this dataset's empirical intercepts). Concern dissipates if anchors generalize to other transformers, items, and samples.
- ▶ Best single-pair anchors must be validated on an independent Big Five sample before being treated as pre-specified.
- ▶ Signed loading recovery requires an explicit polarity correction step — do not present polarity-adjusted  $r$  as raw signed recovery.
- ▶ Absolute all-item loading recovery = domain salience recovery, not full signed psychometric loading recovery.
- ▶ Average wrong-axis baseline (.560) is non-trivial. For Agreeableness, correct-axis semantic location (.741) does not beat average wrong-axis (.779) but does beat single wrong-axis (.617) — the more appropriate comparison when average wrong-axis is G-contaminated.
- ▶ Centroid axes are vulnerable to evaluative/arousal bleeding. Use single-pair anchors or validate that centroid axes are not G-dominated.
- ▶ Anchor directions may not generalize across embedding models. Cross-model validation is needed before anchors are treated as fixed.
- ▶ All  $r$  are LOO averages (10 items, LOO  $n = 9$ ).
- ▶ When items trigger response processes beyond literal meaning, semantic and response FA parameters may diverge — a process to investigate, not a failure.

# Three Temptations to Resist

1. **'Before response data' not 'without response data'.**

We advocate 'before response data' rather than 'without response data', because for some applications quantities derived from human responses are the required criterion. The phrase 'without' overstates independence from empirical validation.

2. **Do not judge PFA by any single recovered quantity.**

PFA should not be judged solely by performance on any single recovered quantity; its value lies in jointly recovering dimensionality, loadings, and semantic locations within one representation.

3. **'Empirical' is not reserved for human responses.**

We do not reserve the term 'empirical' for human response data, recognizing that embedding data themselves are empirical with provenance that is ultimately human.

# References I

- ▶ Guenole, N., Samo, A., & Sun, T. (2024, July 16–19). Pseudo-factor analysis of embedding similarity matrices [Paper presentation]. *International Meeting of the Psychometric Society*, Prague, Czech Republic.
- ▶ Guenole, N., D'Urso, E. D., Samo, A., Sun, T., & Haslbeck, J. M. (2025). Enhancing scale development: Pseudo-factor analysis of language embedding similarity matrices. *PsyArXiv*.
- ▶ Guenole, N. (2026). Pseudo-Factor Analysis with Item Intercepts. *OSF Preprint*.
- ▶ Guenole, N. (no date). Psychometrics.ai: Transforming Behavioral Science with Machine Learning. Available at: <https://psychometrics.ai> (Accessed: date of access).
- ▶ Grand, G., Blank, I. A., Pereira, F., & Fedorenko, E. (2022). Semantic projection recovers rich human knowledge of multiple object features from word embeddings. *Nature Human Behaviour*, 6(7), 975–987.
- ▶ Hernandez, I., & Nie, W. (2023). The AI-IP: Minimizing the guesswork of personality scale item development through artificial intelligence. *Personnel Psychology*, 76(4), 1011–1035.
- ▶ Hommel, B. E., & Arslan, R. C. (2025). Language models accurately infer correlations between psychological items and scales from text alone. *Advances in Methods and Practices in Psychological Science*, 8(4).
- ▶ Lorenzo-Seva, U., Timmerman, M. E., & Kiers, H. A. (2011). The Hull method for selecting the number of common factors. *Multivariate Behavioral Research*, 46(2), 340–364.
- ▶ Milano, N., Luongo, M., Ponticorvo, M., & Marocco, D. (2025). Semantic analysis of test items through large language model embeddings predicts a-priori factorial structure of personality tests. *Current Research in Behavioral Sciences*, 8, 100168.
- ▶ Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. arXiv:1908.10084.
- ▶ Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- ▶ Russell-Lasalandra, L., Golino, H., Garrido, L. E., & Christensen, A. P. (2026). The ultimate tutorial for AI-driven scale development in generative psychometrics: Releasing AIGENIE from its bottle. *arXiv preprint arXiv:2603.28643*.

## References II

- ▶ Sörbom, D. (1974). A general method for studying differences in factor means and factor structure between groups. *British Journal of Mathematical and Statistical Psychology*, 27(2), 229–239.
- ▶ Suárez-Álvarez, J., He, Q., Guenole, N., & D'Urso, D. (2026). Using artificial intelligence in test construction: A practical guide. *Psicothema*, 38(1), 1–12.
- ▶ Varrasi, S., Platania, G. A., Castellano, S., Ferraioli, F., Massimino, S., Vicario, C. M., & D'Urso, E. D. (2026). Expanding psychometrics with pretrained language models: Evaluating pseudo-factor analysis in applied and multilingual contexts. *Methods in Psychology*, 100244.
- ▶ Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.
- ▶ Wang, L., Yang, N., Huang, X., Jiao, B., Yang, L., Jiang, D., Majumder, R., & Wei, F. (2022). Text embeddings by weakly-supervised contrastive pre-training. *arXiv preprint arXiv:2212.03533*.
- ▶ Xiao, S., Liu, Z., Zhang, P., & Muennighoff, N. (2023). C-Pack: Packaged resources to advance general Chinese embedding. *arXiv preprint arXiv:2309.07597*.